Guidance and Control of Unmanned Autonomous Vehicles

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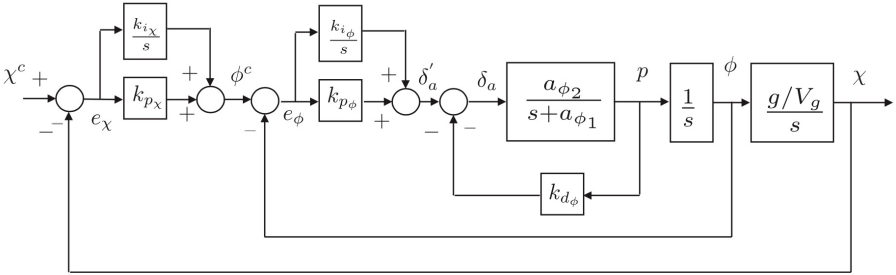
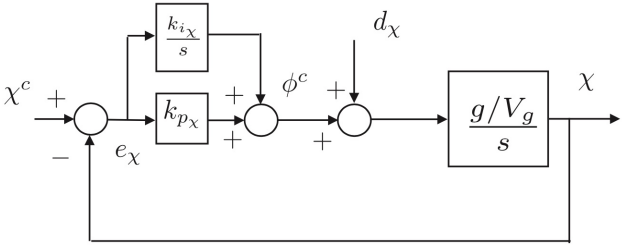
UAV Autopilot: Course Hold





UAV Lateral Autopilot: Course Hold

- With proper tuning of gains, $H_{\phi/\phi^c} \approx 1$ over frequencies between 0 to ω_{n_ϕ}
- Objective: Design k_{p_χ} and k_{i_χ} s.t. $\chi \to \chi^c$ asymptotically
- Transfer functions relating χ, χ^c, and d_χ

$$\chi = \frac{k_{p_\chi}\frac{g}{V_g}s + k_{i_x}\frac{g}{V_g}}{s^2 + k_{p_\chi}\frac{g}{V_g}s + k_{i_\chi}\frac{g}{V_g}}\chi^c + \frac{\frac{g}{V_g}s}{s^2 + k_{p_\chi}\frac{g}{V_g}s + k_{i_\chi}\frac{g}{V_g}}d_\chi$$

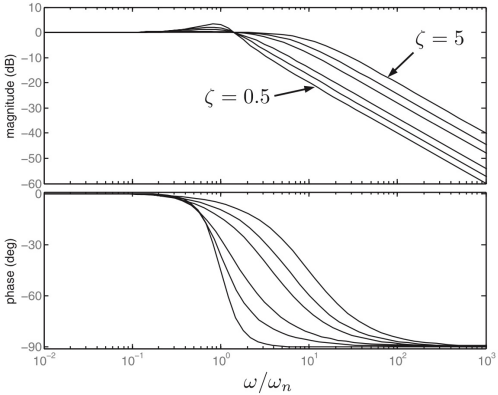
- Using the final value theorem, with constant d_χ and χ^c, $\chi \to \chi^c$
- Desired transfer function from χ^c to χ

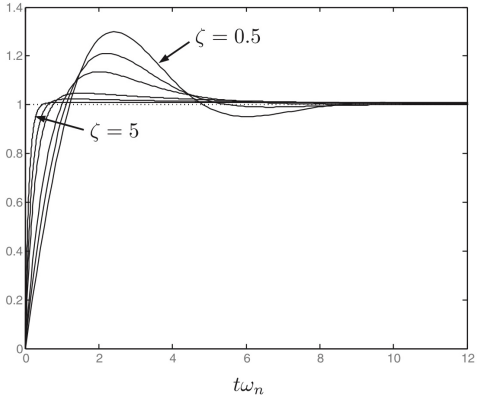
$$H_\chi = \frac{2\zeta_\chi\omega_{n_\chi}s + \omega_{n_\chi}^2}{s^2 + 2\zeta_\chi\omega_{n_\chi}s + \omega_{n_\chi}^2}$$

- **Gains selection**: Based on desired ζ_χ and ω_{n_χ}

UAV Autopilot: Course Hold, Selection of ω_{n_χ} and ζ_χ

- Due to zero in numerator, a standard intuition for selection of ζ does not hold.





- Larger ζ results in larger bandwidth and smaller overshoot.

UAV Lateral Autopilot: Course Hold

- On comparing the coefficients, we get

$$\omega_{n_\chi}^2 = \frac{k_{i_\chi} g}{V_g} \implies \boxed{k_{i_\chi} = \frac{\omega_{n_\chi}^2 V_g}{g}}, \quad 2\zeta_\chi \omega_{n_\chi} = \frac{k_{p_\chi} g}{V_g} \implies \boxed{k_{p_\chi} = \frac{2\zeta_\chi \omega_{n_\chi} V_g}{g}}$$

- To ensure a sufficient bandwidth gap

$$\boxed{\omega_{n_\chi} = \frac{\omega_{n_\phi}}{W_\chi}, \quad W_\chi \geq 5}$$

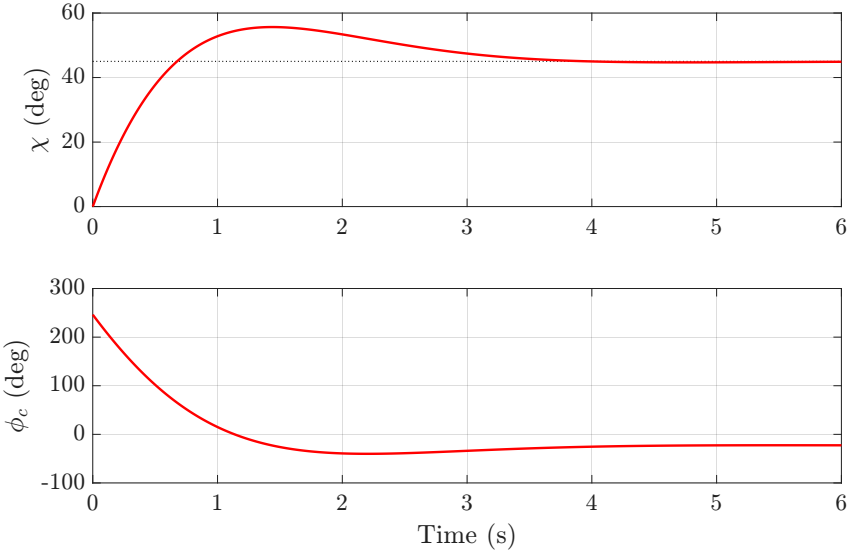
- Large bandwidth separation is better, How can this be ensured?

▷ Slower response of χ-loop, lower ω_{n_χ} ▷ Faster response of ϕ-loop, higher ω_{n_ϕ}

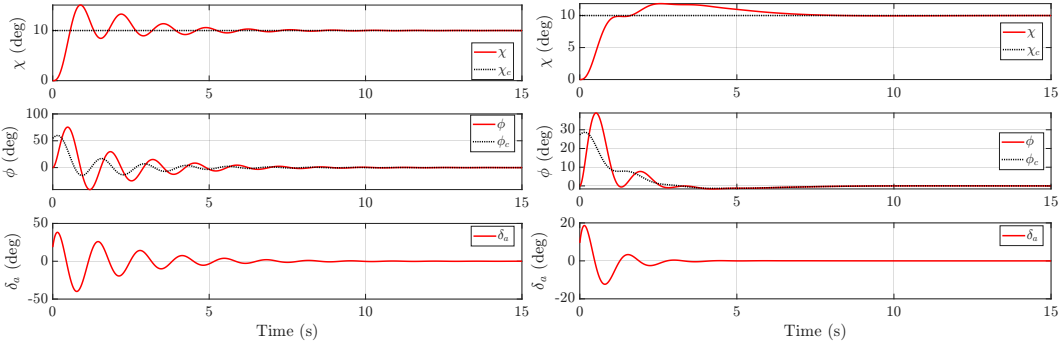
- **More control demand for a faster response**

$$\boxed{\phi^c = k_{p_\chi}(\chi^c - \chi) + \frac{k_{i_\chi}}{s}(\chi^c - \chi)}$$

UAV Lateral Autopilot: Course Hold



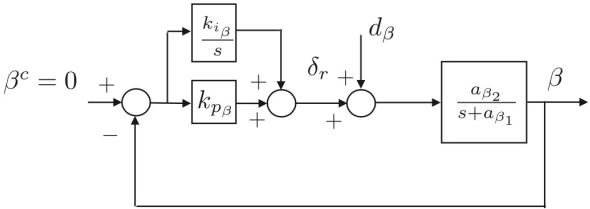
UAV Lateral Autopilot: Integrated Course and Roll



Effect of bandwidth separation (5 and 10 times)

UAV Lateral Autopilot: Sideslip Hold

- Can we maintain a constant or zero sideslip angle?
- Yes, using the rudder inputs



- Transfer function from β^c to β

$$H_{\beta/\beta^c}(s) = \frac{a_{\beta_2}(k_{p_\beta}s + k_{i_\beta})}{s^2 + (a_{\beta_1} + a_{\beta_2}k_{p_\beta})s + a_{\beta_2}k_{i_\beta}}, \;\; H_{\beta/\beta^c}(0) = 1$$

- Characteristic equation for desired closed-loop poles

$$s^2 + 2\zeta_\beta\omega_{n_\beta}s + \omega_{n_\beta}^2 = 0$$

UAV Lateral Autopilot: Sideslip Hold

- On equating the coefficients of characteristic equations

$$2\zeta_\beta \omega_{n_\beta} = a_{\beta_1} + a_{\beta_2} k_{p_\beta}, \quad \omega_{n_\beta}^2 = a_{\beta_2} k_{i_\beta}$$

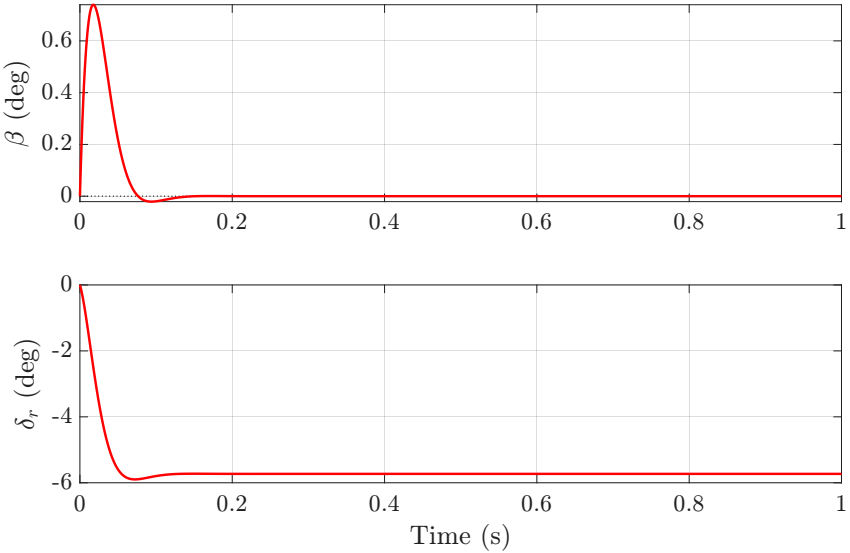
- Maximum error in sideslip ($e_\beta^{\max}$) and maximum allowable rudder deflection ($\delta_r^{\max}$)
- Gains with such constraints

$$k_{p_\beta} = \frac{\delta_r^{\max}}{e_\beta^{\max}} \text{sign}(a_{\beta_2}), \; k_{i_\beta} = \frac{\omega_{n_\beta}^2}{a_{\beta_2}} = \frac{1}{a_{\beta_2}} \left(\frac{a_{\beta_1} + a_{\beta_2} k_{p_\beta}}{2\zeta_\beta} \right)^2$$

- Output of sideslip loop (rudder deflection)

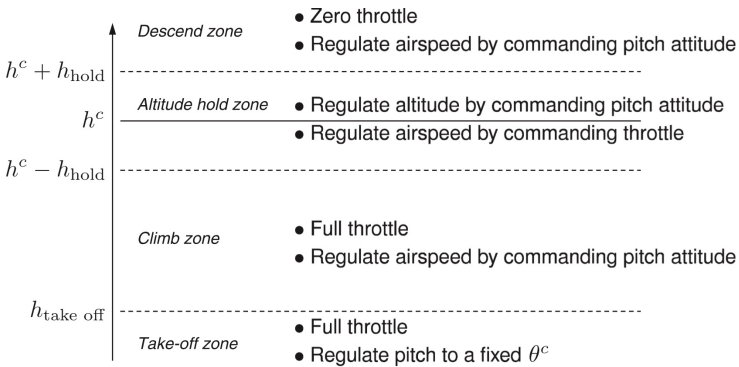
$$\boxed{\delta_r = -k_{p_\beta} \beta - \frac{k_{i_\beta}}{s} \beta}$$

UAV Lateral Autopilot: Sideslip Hold



UAV Longitudinal Autopilot

- Complicated longitudinal autopilot as compared to the lateral ones
- Role of airspeed in the longitudinal dynamics
- **Objective**: Regulate airspeed and altitude using throttle and elevator inputs
- **Regulation of altitude and airspeed**: Role of altitude error



UAV Longitudinal Autopilot

- **Take-off zone**: Full throttle is commanded and $\theta \to \theta^c$ using the elevator.
- **Climb zone**: To maximize the climb rate
  - To maximize the climb rate, full throttle is commanded, and airspeed is regulated using pitch angle. How?
  - If airspeed increases above the nominal value, then the aircraft is caused to pitch up, leading to an increase in climb rate and a decrease in airspeed.
  - If the airspeed drops below the nominal value, the aircraft is pitched down, increasing airspeed but also with a decrease in climb rate.
  - Regulating airspeed using pitch attitude effectively avoids stall conditions.
  - No regulation of airspeed with pitch attitude immediately after the take-off

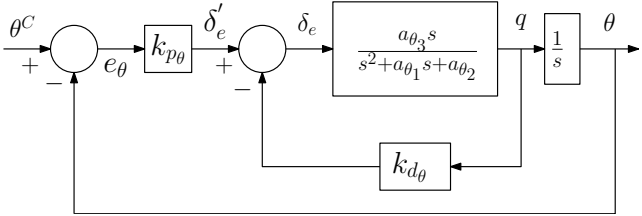
UAV Longitudinal Autopilot

- **Descend zone**: Similar to climb zone, except a zero throttle
  - Avoidance of stall conditions by regulating airspeed using pitch angle
  - Maximization of descent rate at a given airspeed
- **Altitude hold zone**
  - **Airspeed**: By adjusting throttle
  - **Altitude**: By commanding pitch attitude
- **Longitudinal autopilot**: Necessary feedback loops

- Pitch attitude hold using elevator
- Altitude hold using pitch attitude
- Airspeed hold using pitch attitude
- Airspeed hold using throttle

UAV Longitudinal Autopilot: Pitch Attitude Hold

- **Pitch attitude hold loop**: Similar to roll attitude hold loop



$$H_{\theta/\theta^c}(s) = \frac{k_{p_\theta} a_{\theta_3}}{s^2 + (a_{\theta_1} + k_{d_\theta} a_{\theta_3})s + (a_{\theta_2} + k_{p_\theta} a_{\theta_3})}$$

- What about the DC gain of this transfer function? Nonunity DC gain
- Desired canonical second-order transfer function: $\dfrac{k_{\theta_{DC}} \omega_{n_\theta}^2}{s^2 + 2\zeta_\theta \omega_{n_\theta} s + \omega_{n_\theta}^2}$

UAV Longitudinal Autopilot: Pitch Attitude Hold

- Equating coefficients of the characteristic equation

$$\omega_{n_\theta}^2 = a_{\theta_2} + k_{p_\theta} a_{\theta_3}, \quad 2\zeta_\theta \omega_{n_\theta} = a_{\theta_1} + k_{d_\theta} a_{\theta_3}$$

- Gain k_{p_θ} chosen to satisfy the actuator constraint, then

$$\boxed{k_{p_\theta} = \frac{\delta_e^{\max}}{e_\theta^{\max}} \text{sign}(a_{\theta_3})}, \qquad \boxed{k_{p_\theta} = \frac{\omega_{n_\theta}^2 - a_{\theta_2}}{a_{\theta_3}} \quad \text{for unbounded } \delta_e}$$

- Consideration of sign of a_{θ_3} that depends on $C_{m_{\delta e}}$ (typically <0)
- Bandwidth limit of the pitch loop:

$$\omega_{n_\theta} = \sqrt{a_{\theta_2} + \frac{\delta_e^{\max}}{e_\theta^{\max}} |a_{\theta_3}|}$$

- Selection of gain k_{d_θ}:

$$k_{d_\theta} = \frac{2\zeta_\theta \omega_{n_\theta} - a_{\theta_1}}{a_{\theta_3}}$$

UAV Longitudinal Autopilot: Pitch Attitude Hold

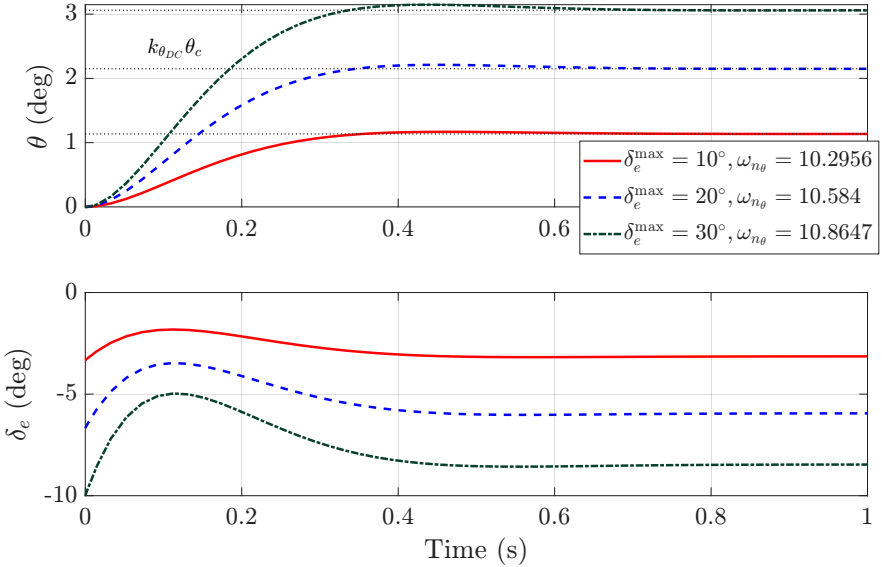
- What about the DC gain of the transfer function $H_{\theta/\theta^c}(s)$ as $k_{p_\theta} \to \infty$?

$$H_{\theta/\theta^c}(s=0) = k_{\theta_{DC}} = \frac{k_{p_\theta} a_{\theta_3}}{a_{\theta_2} + k_{p_\theta} a_{\theta_3}} \to 1$$

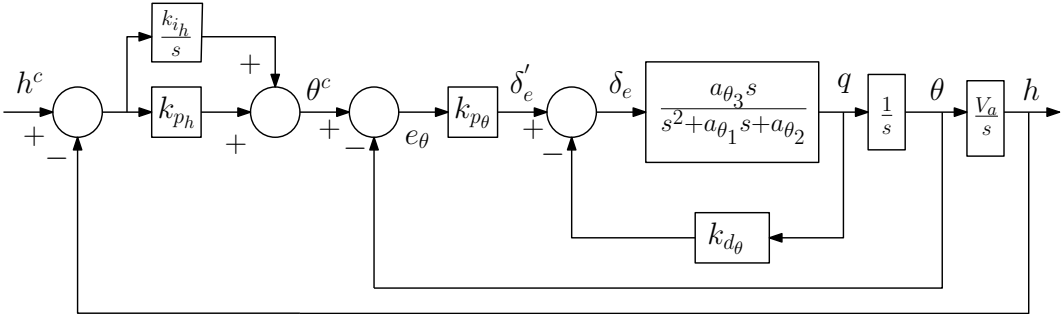
- What about the DC gain value with a nominal value of k_{p_θ}?
- Significantly less than one, for typical gain values of k_{p_θ}
- Can we obtain unity DC gain for this transfer by modifying the controller?
- **Integral feedback to ensure unit DC gain**
- **Drawback**: Limitation on bandwidth by use of an integral term
- Elevator deflection input (Output of pitch attitude hold loop)

$$\boxed{\delta_e = k_{p_\theta}(\theta^c - \theta) - k_{d_\theta} q}$$

AV Longitudinal Autopilot: Pitch Attitude Hold



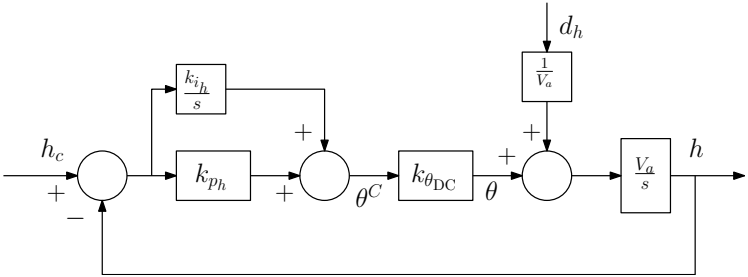
UAV Longitudinal Autopilot: Altitude Hold using Pitch



- How to deal with the attitude hold design?
- Can we replace the inner loop with unity, as before? No Why?
- With a DC gain of $k_{\theta_{DC}}$,

$$\boxed{\theta = k_{\theta_{DC}} \theta^c}$$

UAV Longitudinal Autopilot: Altitude Hold using Pitch



$$h(s) = \left[\frac{k_{\theta_{DC}} V_a k_{p_h} \left(s + k_{i_h}/k_{p_h}\right)}{s^2 + k_{\theta_{DC}} V_a k_{p_h} s + k_{\theta_{DC}} V_a k_{i_h}}\right] h^c(s) + \left[\frac{s}{s^2 + k_{\theta_{DC}} V_a k_{p_h} s + k_{\theta_{DC}} V_a k_{i_h}}\right] d_h(s)$$

- What about the DC gain and the disturbance rejection?
- DC gain of one and rejection of a constant disturbance input
- Independent closed-loop transfer function on vehicle parameters, except V_a

UAV Longitudinal Autopilot: Altitude Hold using Pitch

- Selection of gains k_{p_h} and k_{i_h} to ensure $\omega_{n_h} = \dfrac{\omega_{n_\theta}}{W_h},\ W_h \in [5 \quad 15]$
- Desired closed-loop transfer function $\dfrac{\omega_{n_h}^2}{s^2 + 2\zeta_{n_h}\omega_{n_h}s + \omega_{n_h}^2}$
- Equating coefficients

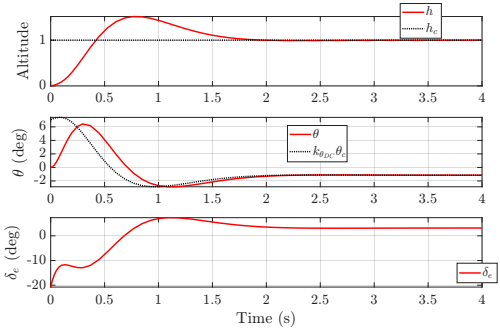
$$2\zeta_{n_h}\omega_{n_h} = k_{\theta_{DC}}V_a k_{p_h} \implies k_{p_h} = \frac{2\zeta_{n_h}\omega_{n_h}}{k_{\theta_{DC}}V_a}$$

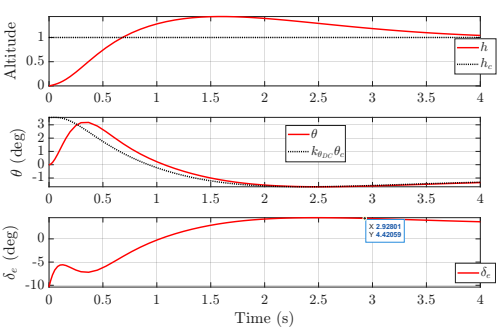
$$\omega_{n_h}^2 = k_{\theta_{DC}}V_a k_{i_h} \implies k_{i_h} = \frac{\omega_{n_h}^2}{k_{\theta_{DC}}V_a}$$

- Pitch command inputs (output of altitude-hold with pitch loop)

$$\boxed{\theta^c = k_{p_h}(h^c - h) + \frac{k_{i_h}}{s}(h^c - h)}$$

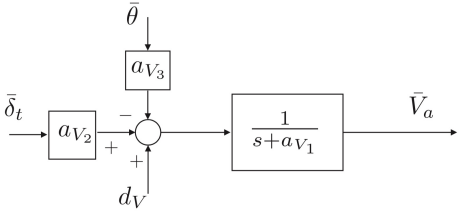
UAV Longitudinal Autopilot: Altitude Hold using Pitch



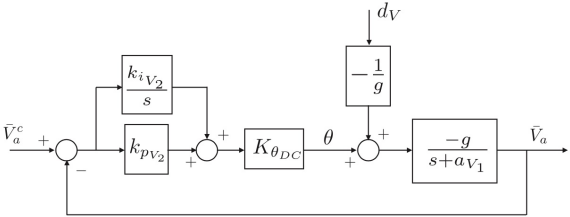


Bandwidth separation of 5 and 10 times

UAV Longitudinal Autopilot: Airspeed Hold using Commanded Pitch



- Disturbance rejection requires an integrator



UAV Longitudinal Autopilot: Airspeed Hold using Commanded Pitch

- Transfer function in the Laplace domain

$$\bar{V}_a(s) = -\left[\frac{k_{\theta_{DC}} g k_{p_{V_2}} \left(s + k_{i_{V_2}}/k_{p_{V_2}}\right)}{s^2 + \left(a_{V_1} - k_{\theta_{DC}} g k_{p_{V_2}}\right) s - k_{\theta_{DC}} g k_{i_{V_2}}}\right] \bar{V}_a^c(s)$$
$$+ \left[\frac{s}{s^2 + \left(a_{V_1} - k_{\theta_{DC}} g k_{p_{V_2}}\right) s - k_{\theta_{DC}} g k_{i_{V_2}}}\right] d_V(s)$$

- What about the DC gain and the disturbance rejection?
- A DC gain of 1 and disturbance rejection
- To hold constant airspeed, $\theta \to \alpha \neq 0$.
- How will this constant pitch angle be generated?
- Integrator will wind up to provide a command value of proper α
- Bandwidth requirement: $\omega_{n_{V_2}} = \dfrac{\omega_{n_\theta}}{W_{V_2}}, \quad W_{V_2} > 5$

UAV Longitudinal Autopilot: Airspeed Hold using Commanded Pitch

- How to obtain the controller gains?
- Using the placement of closed-loop poles at desired locations

$$s^2 + 2\zeta_{V_2}\omega_{n_{V_2}} s + \omega_{n_{V_2}}^2 = 0$$

- Equating coefficients of the characteristic equation

$$\omega_{n_{V_2}}^2 = -k_{\theta_{DC}} g k_{i_{V_2}}, \quad 2\zeta_{V_2}\omega_{n_{V_2}} = a_{V_1} - k_{\theta_{DC}} g k_{p_{V_2}}$$

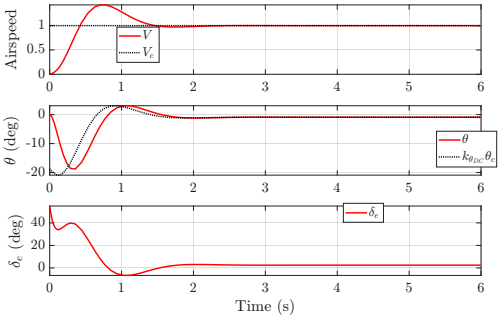
- Gains selection

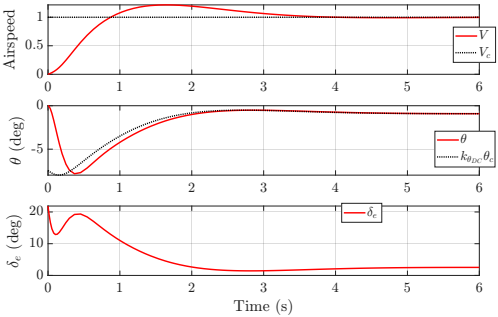
$$\boxed{k_{i_{V_2}} = -\frac{\omega_{n_{V_2}}^2}{k_{\theta_{DC}} g}, \quad k_{p_{V_2}} = \frac{a_{V_1} - 2\zeta_{V_2}\omega_{n_{V_2}}}{k_{\theta_{DC}} g}}$$

- Pitch angle command (output of airspeed hold using pitch loop)

$$\boxed{\theta^c = k_{p_{V_2}} \left(V_a^c - V_a\right) + \frac{k_{i_{V_2}}}{s} \left(V_a^c - V_a\right)}$$

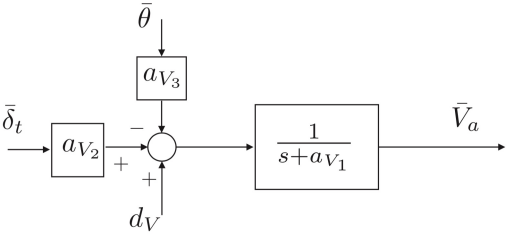
UAV Longitudinal Autopilot: Airspeed Hold using Commanded Pitch

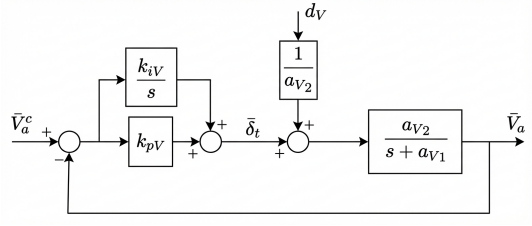




Bandwidth separation of 4 and 8 times

UAV Longitudinal Autopilot: Airspeed Hold using Throttle





- Transfer function with proportional (P) control

$$\bar{V}_a(s) = \left[\frac{a_{V_2}k_{pV}}{s+(a_{V_1}+a_{V_2}k_{pV})}\right]\bar{V}_a^c(s) + \left[\frac{1}{s+(a_{V_1}+a_{V_2}k_{pV})}\right]d_V(s)$$

- Nonunity DC gain and no step disturbance rejection
- What do you expect with the proportional and integral (PI) type feedback?

UAV Longitudinal Autopilot: Airspeed Hold using Throttle

- Transfer function with PI feedback control

$$\bar{V}_a(s) = \left[\frac{a_{V_2}\left(k_{p_V} s + k_{i_V}\right)}{s^2 + \left(a_{V_1} + a_{V_2} k_{p_V}\right) s + a_{V_2} k_{i_V}}\right] \bar{V}_a^c(s) + \left[\frac{s}{s^2 + \left(a_{V_1} + a_{V_2} k_{p_V}\right) s + a_{V_2} k_{i_V}}\right] d_V(s)$$

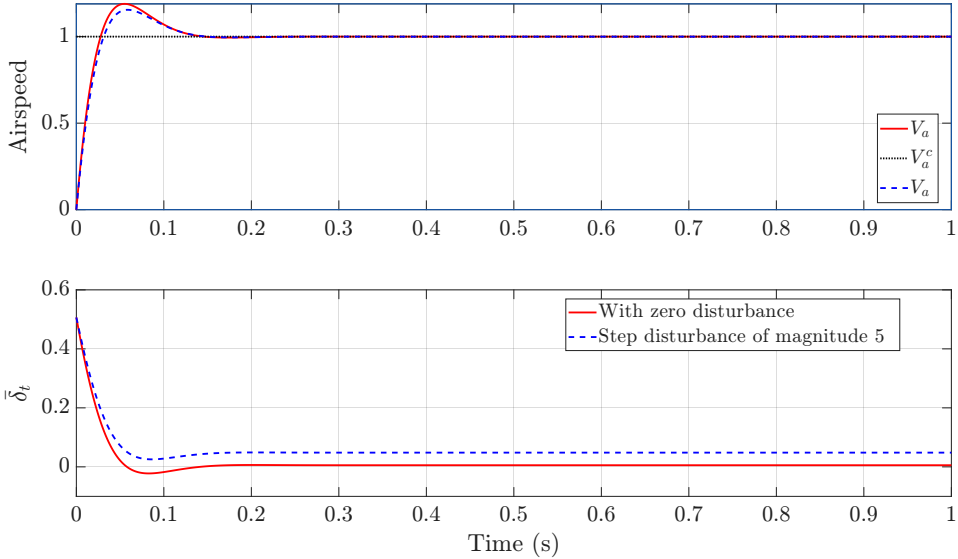
- Unity DC gain and step disturbance rejection
- Selection of gains using the pole placement approach
- Equating coefficients: $\boxed{2\zeta_V \omega_{n_V} = a_{V_1} + a_{V_2} k_{p_V}, \quad \omega_{n_V}^2 = a_{V_2} k_{p_V}}$
- Controller gains selection

$$\boxed{k_{i_V} = \frac{\omega_{n_V}^2}{a_{V_2}}, \quad k_{p_V} = \frac{2\zeta_V \omega_{n_V} - a_{V_1}}{a_{V_2}}}$$

- Throttle input command

$$\boxed{\bar{\delta}_t = k_{p_V}\left(\bar{V}_a^c - \bar{V}_a\right) + \frac{k_{i_V}}{s}\left(\bar{V}_a^c - \bar{V}_a\right)}$$

UAV Longitudinal Autopilot: Airspeed Hold using Throttle



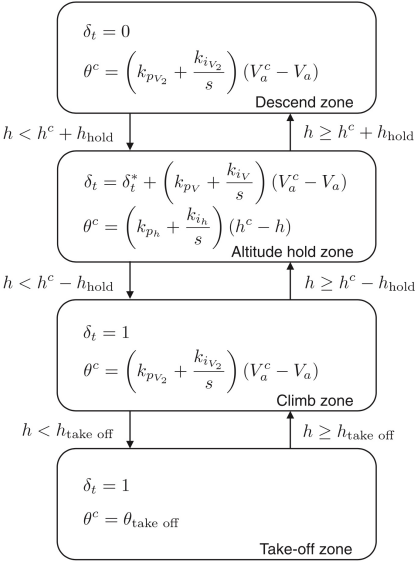
UAV Longitudinal Autopilot: Airspeed Hold using Throttle and State Machine

- Can we implement the controller without the knowledge of trim velocity?
- Yes, fortunately, we can do so. How?
- As $\bar{V}_a^c = V_a^c - V_a^\star$ and $\bar{V}_a = V_a - V_a^\star$, error signal

$$\boxed{e = \bar{V}_a^c - \bar{V}_a = V_a^c - V_a}$$

- **Throttle command**: $\delta_t = \delta_t^\star + \bar{\delta}_t$
- Throttle input (output of airspeed hold):

$$\boxed{\delta_t = \delta_t^\star + k_{p_V}\left(V_a^c - V_a\right) + \frac{k_{i_V}}{s}\left(V_a^c - V_a\right)}$$



UAV Longitudinal Autopilot: Altitude Control State Machine

- **Longitudinal autopilot**: Control of motions in body $\boldsymbol{i}^b$-$\boldsymbol{k}^b$ plane
- Four basic designs together complete overall longitudinal autopilot
- **Climb zone**: $\delta_t = 1$ and airspeed hold from commanded pitch mode to control airspeed, thus avoiding stall conditions.
- Vehicle climbs at its **maximum possible climb rate** until it is close to the altitude set point.
- **Descend zone**: $\delta_t = 0$ and airspeed hold from commanded pitch mode.
- Descend at a steady rate until the vehicle reaches the altitude hold zone.
- **Altitude hold zone**: Airspeed-from-throttle mode to regulate airspeed.
- Altitude-from-pitch mode is used to regulate altitude around h^c.
- **Pitch attitude control loop is active in all four zones.**

Textbooks/References

References

- Randal Beard and Timothy W. McLain, "Small Unmanned Aircraft: Theory and Practice", Princeton University Press, 2012

Thank you for your attention !!!